

Instructions

Identify design requirements for RHMS: main components, user needs, expected behaviors, and first design decisions. Produce a short document and a conceptual diagram.

Remote Health Monitoring System Design REquirement Document

Main Components

The main component consists of an IoT device, network for connectivity, cloud server, database, hospital dashboard, doctor's dashboard, patient's dashboard, and alert system. The patient wears the IoT device and it automatically collect vital informations such as blood pressure, heart rate, and temperature. It then send this information to the server. The network which in our case is the internet, enables the IoT device to transmit data, the cloud server receives, stores, and process patient's health data, the database is where patients, doctor, and hospital data will be stored, the hospital dashboard enables the hospital to monitor patient's health data and doctor's data, the doctor's dashboard is a platform where doctor receives patient's report such as abnormal readings and also a platform for the doctor to send feedback, the patient dashboard is where patient recieves doctor's feedback, and the alert system notifies the doctor as soon as abnormal reading is recorded by the IoT device.

User Need

The patient's want to wear a convenient device that automatically record his/her health data and send it to the doctor if abnormality is detected.

Doctor want to receive alert with patient's health data if the readings are abnormal as quick as possible, send feedback as fast as possible and also be able to view health history.

Hospital Administration want to manage doctor and patient information and monitor the overall system performance.

Expected Behavior

The IoT device collect's patient's health data and transmit it periodically, the system analyze the data and check if there is abnormality. If abnormality is detected, the system generate alert and send it to the doctor. The doctor review and analyze the data and provide recommendation, and the system forward this feedback to the patient.

First Design Decision

The system will use client-server architecture with cloud data storage. The IoT device will send data using MQTT(Message Queuing Telemetry Transport) protocol. The programming language to be use is java and python, encryption and authentication will be use for security.

